

Subsystem	Component	Target	Position of Axis to Align	Axis and Frame to Align	Required Accuracy	Required Precision	Function / Motivating Requirement
Payload	GPR sounder antenna (REASON-heritage)	Nadir / local surface normal	Belly-mounted, phase center 0.35 m above the ice, clear of wheels and jettisoned struts	Antenna boresight (+Z body, down) aligned to the local vertical	+5 deg of nadir (3 sigma)	+0.5 deg stop-to-stop repeatability	Pointing. Bounds geometric error in the two-way travel time (REQ. 03.02)
Payload	GPR trace georeference	Local ENU frame origin set at the landing site	Trace tagged at the wheel-contact centroid, antenna offset resolved out Inside the radiation vault, no more than 0.05 m from the visual odometry camera frame origin	Antenna phase center offset rotated from body into local ENU	+0.5 m per 50 m traverse segment	+0.1 m stop-to-stop	Localization. Places each trace on a continuous along-track profile (REQ. 02.01, REQ.03.01)
GNC	MEMS IMU (2x, cross-strapped)	N/A (inertial reference)		Sensitive axes aligned to body XYZ within 0.1 deg after calibration	Roll/pitch +0.5 deg (3 sigma); bias stability 1.5 deg/h or better	0.05 deg arc-to-arc	Attitude determination and inertial propagation between heading fixes (REQ.11.01)
GNC	Fine sun sensor (2x)	Sun	Deck-mounted on opposite corners, at least 60 deg of clear sky away from the mast and helix	Sensor boresight to deck normal (+Z body); Sun vector resolved to azimuth through the ephemeris and clock	+0.1 deg Sun vector; +0.25 deg heading (3 sigma) after ephemeris and alignment error	+0.05 deg	Absolute heading. The only non-inertial azimuth reference available in a GPS-denied environment with no usable magnetic
GNC	Stereo navigation cameras	Terrain 1.5 m to 8 m ahead of the front wheels	Mast-mounted 1.2 m above the deck, 0.20 m stereo baseline	Boresight 25 deg below the local horizontal, pan +-180 deg about the mast axis	0.02 m range error at 3 m; detects a 0.20 m step at 8 m	+1 pixel disparity	Visual odometry and hazard assessment (REQ.11.02, REQ.11.03)
GNC	Onboard clock and timebase	N/A	Vault-mounted, 1 PPS distributed to payload, GNC, and comms	N/A (time reference only)	+1 s absolute over a sol, which bounds the Sun-vector azimuth contribution to 0.02 deg	+10 ms trace-to-trace	Ephemeris lookup for the Sun sensor and time alignment of radar traces to position fixes
GNC	Orbital terrain map registration	Orbital relay terrain map frame	Local stereo terrain patch correlated against the consensus map	Local ENU frame rotated into the orbital map frame	+10 m absolute in the map frame	+2 m re-registration repeatability	Localization recovery after a navigation hold (REQ. 11.04, REQ.01.03)
Mobility	Drive actuators with wheel encoders (4x)	Commanded drive arc	One per corner on the rocker, 0.32 m wheel diameter	Wheel rotation axis normal to the body XZ plane within 0.5 deg	+2% of commanded arc length before slip correction	+0.02 m per 5 m arc	Trajectory following, and the slip observable against visual odometry (REQ.11.03)
Mobility	Rocker suspension and chassis attitude	Local surface plane	Single pivot on the chassis belly centerline	Body Z axis to the local vertical	Static tilt knowledge +0.5 deg	+0.2 deg	Holds the sounder inside the nadir budget on uneven ice (REQ.03.02)
Mobility	Mast pan and tilt actuator	Next planned drive arc	Mast head, 1.2 m above the deck	Camera boresight azimuth to the planned arc heading in the body frame	+2 deg of the commanded arc heading	+0.5 deg	Points the stereo pair at the corridor that is about to be driven (REQ.11.02)
COMMs	UHF quadrifilar helix	Nearer orbital relay agent during a pass	Mast top, phase center at least 1.0 m above the deck with a clear hemisphere	Helix boresight to the local zenith (+Z), body-fixed, no mechanism	No active pointing required; +60 deg off-boresight usable at the 10 deg elevation mask	N/A (no mechanism)	Link geometry. The hemispherical pattern is what removes a pointing requirement (REQ.10.03)
COMMs	X-band LGA (contingency beacon)	Earth	Deck edge, unobstructed toward the ecliptic	LGA boresight inside the 8 dBi beam toward the Earth line of sight	+20 deg of the Earth line of sight	N/A	Contingency direct-to-Earth beacon pointing, the 5 bps fallback floor in REQ.10.03
Thermal	Radiator and heat-pipe panels	Cold sky	Deck top, shaded from the RTG and from the disc of Jupiter	Panel normal within 30 deg of anti-nadir	+10 deg of the as-designed view factor	N/A	Heat rejection view factor (REQ.09.01)
Structures	Landing struts and footpads (jettisoned after checkout)	Local surface at touchdown	Four footpads on a common plane below the belly	Deck plane to the local horizontal	+5 deg deck level at touchdown	+2 deg leg-to-leg	Sets the initial sounder standoff and the attitude initialization reference (REQ.07.01)